Accelerated T2 Brain MRI Reconstruction with Deep Learning: A Multi-Stage Study on BraTS

唐志昊 (2022533131) 应弘 (2023531009)

BME AI Project 1 — Group 4

Config: `config/formal_train.yaml` Pipeline run: 2026-04-28 15:21:32

Abstract

Accelerated magnetic resonance imaging (MRI) reconstruction from undersampled k-space is a core problem in parallel and compressed-sensing MRI. This work evaluates a three-stage pipeline on the BraTS glioma dataset: (i) simulation of random variable-density undersampling at acceleration factor AF$=5$; (ii) a supervised U-Net baseline mapping aliased T2 to fully sampled T2; and (iii) a multi-modal unrolled U-Net with explicit k-space data consistency, using undersampled T2 jointly with fully sampled T1. Patient-level splitting avoids leakage across slices from the same subject; intensity normalization applies z-score scaling on non-zero voxels. Quantitatively, the baseline improves peak signal-to-noise ratio (PSNR) from 25.76 dB to 35.18 dB and structural similarity (SSIM) from 0.3628 to 0.9271; the unrolled multi-modal model reaches 38.12 dB PSNR and 0.9660 SSIM, exceeding the baseline by 2.93 dB PSNR and 0.0389 SSIM. Results support that anatomical guidance from T1 together with data-consistency unrolling yields stronger fidelity than single-modality CNN reconstruction alone.

Index Terms—MRI reconstruction, accelerated imaging, BraTS, deep learning, U-Net, unrolled optimization, data consistency, undersampled k-space.

1 Introduction

Accelerating MRI acquisitions by undersampling k-space reduces scan time but introduces aliasing when reconstructions neglect sampling patterns; learning-based reconstruction has become a standard complement to analytical parallel-imaging and compressed-sensing pipelines in both research and routine engineering practice. Brain tumor protocols such as BraTS provide multi-modal volumes (including T1 and T2); when T2 k-space alone is undersampled, fully sampled T1 can provide complementary anatomical contrast for learned reconstruction.

This technical report summarizes an implementation-centric study with three aligned objectives: *(i)* synthesize realistic undersampled T2 slices with controlled acceleration; *(ii)* train a strong single-channel baseline (U-Net); and *(iii)* extend to an unrolled architecture with data-consistency layers and T1-conditioned reconstruction. All experiments share patient-level splits and comparable preprocessing.

2 Materials and Methods

2.1 Dataset and preprocessing

We use BraTS-style glioma volumes with T1 and T2 modalities (dataset root configurable; default `../bmeaidataset`). Processing uses axial slices (slice axis = 2): z-score normalization on non-zero voxels only; patient-level train/validation/test partitioning so adjacent slices from one subject do not leak across splits. The formal run retains *16 central slices per patient*; slices are cached in RAM where feasible to reduce I/O on consumer GPUs (RTX 4060 laptop, RTX 5070Ti desktop). Aggregated slice counts are summarized in Table 1.

Table 1: Slice counts after patient-level splitting (formal configuration).

Split	Slices
Train	6992
Validation	1504
Test	1504

2.2 Undersampling simulation (Task 1)

A two-dimensional random variable-density sampling mask with acceleration AF$= 5$ is drawn in k-space. Specifically, k-space is commonly depicted on a rectangular grid with principal axes k_x and k_y, aligned with spatial-frequency content along the image-domain horizontal (x) and vertical (y) directions rather than with anatomical coordinates [1]. Fully sampled T2 slices undergo forward FFT, masking, and inverse

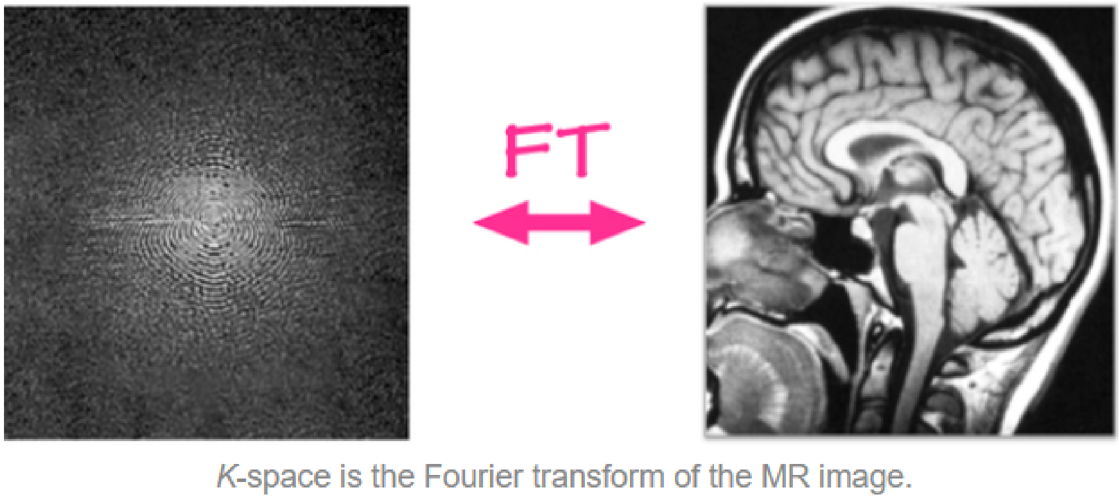

Figure 1: Schematic illustration of k-space layout and axes.

FFT to produce aliased images used as inputs for Tasks 2–3.

2.3 Networks and training

Task 2 (U-Net baseline). A U-Net maps aliased T2 to fully sampled T2 with mean squared error (MSE). Optimization uses Adam-style updates with learning rate 5×10^{-4} and `ReduceLROnPlateau`. Implementation optimizations include pinned CPU tensors, non-blocking host–device copies, `channels_last`, TF32 where applicable, and prefetch-friendly dataloaders.

Task 3 (unrolled multi-modal model). An unrolled U-Net with two cascades integrates data-consistency (DC) layers enforcing consistency with measured undersampled k-space. Inputs concatenate undersampled/reconstructed T2 with fully sampled T1. Training minimizes a hybrid loss (L1 weight 0.7). DC layers encode the sampling operator so outputs remain faithful to acquired frequencies while T1 stabilizes anatomy.

Hyperparameters for the formal comparison are listed in Table 2.

Table 2: Representative hyperparameters for Task 2 vs. Task 3 (formal run).

Setting	Task 2	Task 3
Input	Aliased T2	Aliased T2 + full T1
Backbone	U-Net	Unrolled U-Net + DC
Base channels	24	16
Depth	4	3
Batch size	8	4
Epochs	10	8
Learning rate	5×10^{-4}	10^{-4}
Loss	MSE	hybrid

3 Experimental Results

3.1 Quantitative metrics

Table 3 reports aggregate PSNR and SSIM relative to fully sampled T2; Table 4 summarizes gains over the aliased baseline; Table 5 isolates Task 3 improvements over Task 2.

Table 3: Main quantitative comparison on the test evaluation.

Method	PSNR (dB)	SSIM
Aliased input	25.76	0.3628
Task 2 U-Net baseline	35.18	0.9271
Task 3 unrolled multi-modal	38.12	0.9660

Table 4: Improvement over aliased input.

Method	ΔPSNR (dB)	ΔSSIM
Task 2 baseline	9.43	0.5643
Task 3 multi-modal	12.36	0.6032

3.2 Overview figures

Figures 2–3 summarize dataset partitioning and aggregate metric contrasts when pipeline assets are available.

3.3 Qualitative reconstruction

Figures 4–9 show undersampling visualization, training curves, example reconstructions, and error maps.

3.4 Challenging cases

Even low-performing slices remain markedly improved versus aliasing-only inputs; Table 6 lists the five weakest Task 3 cases by ranking metric.

Table 5: Task 3 vs. Task 2 (same evaluation protocol).

Metric	Value
ΔPSNR	+2.93 dB
ΔSSIM	+0.0389

[Figure missing — expected file:]

`outputs\_formal/report\_assets/dataset\_split.png`

Figure 2: Patient/slice allocation across splits (pipeline asset).

[Figure missing — expected file:]

`outputs\_formal/report\_assets/metric\_comparison.png`

Figure 3: Summary metric comparison across stages.

Table 6: Five lowest-performing Task 3 slices (post-hoc ranking).

Rank	Patient	Slice	PSNR before	PSNR after	SSIM before	SSIM after
1	BraTS-GLI-00077-000	76	23.26	34.69	0.2865	0.9443
2	BraTS-GLI-00077-000	82	23.14	34.74	0.2875	0.9502
3	BraTS-GLI-00088-001	82	24.33	34.82	0.3160	0.9439
4	BraTS-GLI-00077-000	78	23.23	34.82	0.2942	0.9481
5	BraTS-GLI-00077-000	79	23.22	34.84	0.2921	0.9484

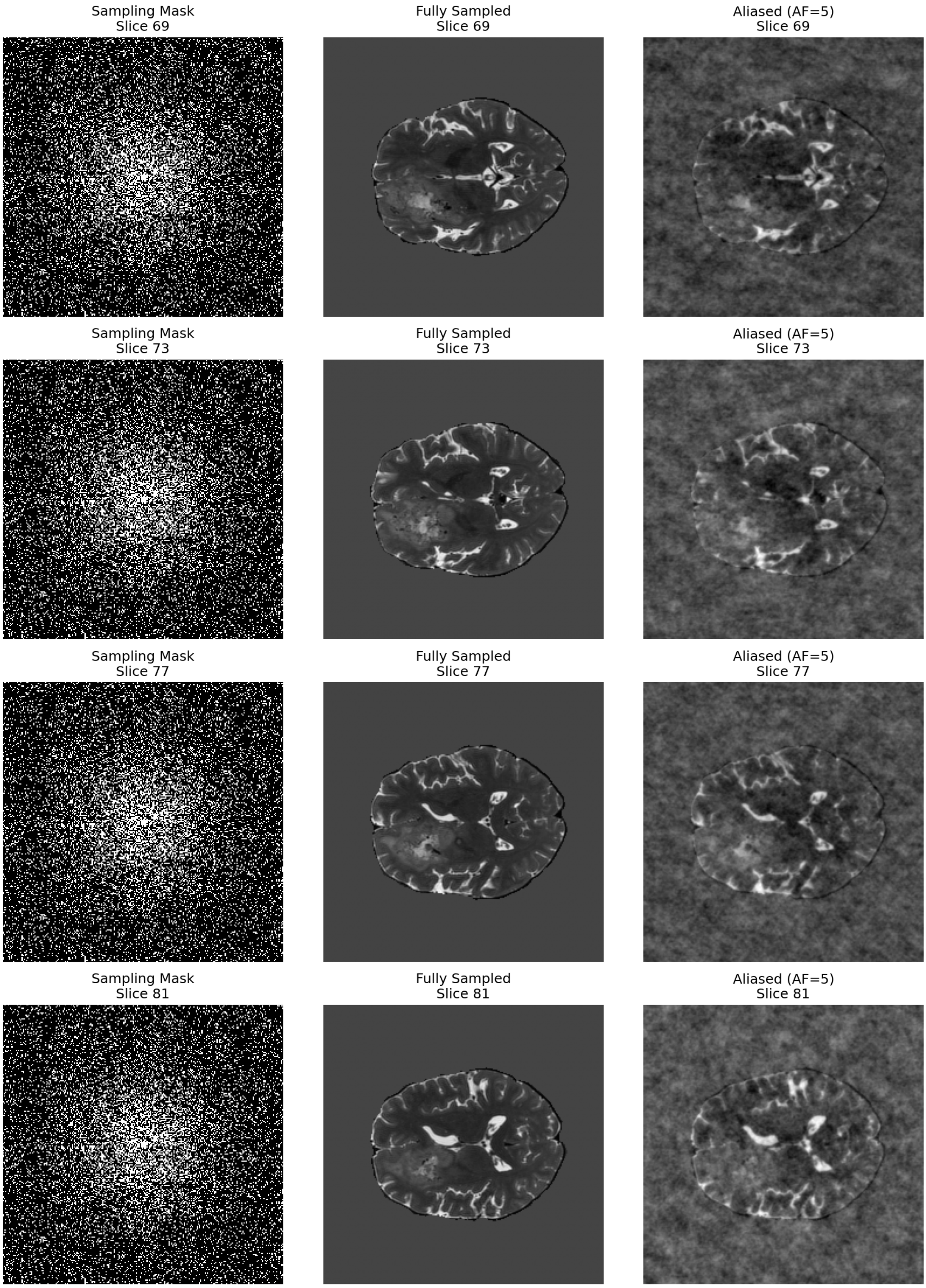

Figure 4: Representative undersampling mask and aliasing behavior (Task 1).

[Figure missing — expected file:]

`outputs\_formal/task2/loss\_curves.png`

Figure 5: Training loss curves for Task 2.

[Figure missing — expected file:]

`outputs\_formal/task2/reconstruction\_results.png`

Figure 6: Qualitative Task 2 reconstructions.

[Figure missing — expected file:]

`outputs\_formal/task3/loss\_curves.png`

Figure 7: Training loss curves for Task 3.

[Figure missing — expected file:]

`outputs\_formal/task3/best\_reconstructions.png`

Figure 8: Representative Task 3 reconstructions.

[Figure missing — expected file:]

`outputs\_formal/task3/error\_analysis.png`

Figure 9: Error analysis for Task 3.

4 Discussion

Across 16 central slices per patient and patient-level splitting, both learned reconstructors substantially outperform the undersampled baseline. Task 2 confirms stable supervised convergence (+9.43 dB PSNR). Task 3 adds multi-modal anatomical guidance and explicit spectral fidelity via DC layers, yielding an additional +2.93 dB PSNR over Task 2 and sharper boundaries in qualitative panels.

Residual failures concentrate on anatomically heterogeneous slices (e.g. complex tumor margins). Potential extensions include edge-preserving or perceptual losses, additional cascades when GPU memory allows, and targeted sampling near challenging anatomy—directions aligned with ongoing work in clinical MRI reconstruction research.

5 Conclusion

We implemented and evaluated a BraTS-aligned pipeline from accelerated k-space simulation through CNN and unrolled multi-modal reconstruction. Quantitative gains and qualitative scans indicate that combining T1 guidance with data-consistent unrolling improves T2 reconstruction fidelity compared with a single-modality U-Net under identical preprocessing and splitting assumptions.

References

[1] MRIQuestions.com, *What is k-space?* Available at `https://mriquestions.com/what-is-k-space.html` (accessed April 29, 2026).

6 Course metadata and reproducibility

Authorship

- Division of labor: 唐志昊 — integration, training experiments, verification, reporting and submission coordination.

AI-assisted tooling

AI tools were used only for coding assistance, experiment orchestration support, and draft editing; methodology choices, validation, interpretation, and submitted artifacts were reviewed manually before submission (presentation slides declare AI assistance per course policy).

Pipeline execution record

- Status: Success.
- Output directory: `outputs_formal`.
- Wall-clock runtime (logged): 24 m 42 s.
- Last completed stage: asset generation.
- Log: `outputs_formal/logs/formal_pipeline.log`.

Submission checklist

- Code repository: prepared.
- Markdown draft: `REPORT.md`.

- LaTeX source: `REPORT.tex`.
- PDF report: `REPORT.pdf`.
- Slides: `slides.tex`, `slides.pdf`.